MANAGING SENTRY: A POINT-OF-SALE AND INVENTORY MANAGEMENT SYSTEM FOR ALEKOS' CORNER GENERAL MERCHANDISE

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TABLE OF CONTENTS

TABLE OF CONTENTS ii

LIST OF FIGURES iv

LIST OF TABLES v

CHAPTER 1 INTRODUCTION 1

1.1 Rationale of the Study 1

1.2 Statement of the Problem 4

1.2.1 General Objective 4

1.3 Significance of the Study 4

1.4 Scope and Limitations 5

CHAPTER 2 RELATED SYSTEMS 7

2.1 Introduction and Theoretical Framework 7

2.2 Intellectual Progression and Thematic Literature Review 9

2.2.1 Evolution from Manual Bookkeeping to Digital Point of Sale in Micro-Retail 10

2.2.2 Inventory Management Models, Automated Restocking, and Discrepancy Control 11

2.2.3 Mobile-First POS Checkout vs. Comprehensive Web Admin Portals 14

2.2.4 Offline-First Database Paradigms and Data Synchronization Architectures 15

2.2.5 Analytical Dashboards, Sales Forecasting, and Operational Decision Support 17

2.3 Review of Related Systems 17

2.3.1 Square POS 18

2.3.2 Loyverse POS 18

2.3.3 Odoo POS (Community Edition) 19

2.3.4 Peddlr 20

2.3.5 Utak POS 20

2.4 Comparative Analysis of Related Systems 21

2.5 Synthesis of Reviewed Literature and Systems 23

CHAPTER 3 DESIGN AND METHODOLOGY 24

3.1 Conceptual Framework 24

3.2 Analysis and Design 28

3.3 Development Model 29

3.4 Development Approaches 29
3.5 Software Development Tools 30
3.6 Project Management 34
3.6.1 Schedule and Timeline 34
3.6.2 Responsibilities 35
3.6.3 Budget and Cost Management 36
3.6.4 Change and Risk Management 36
3.7 Verification, Validation and Testing 39
3.7.1 Verification 39
3.7.2 Validation 41
3.7.3 Functional Testing 42
BIBLIOGRAPHY 45
APPENDICES 48
Appendix A Project Charter 48
Appendix B Transmittal Letter 51
Appendix C Interview Guide 52
Appendix D Questionnaire 53
Appendix E Software Requirement Specifications 54
CURRICULUM VITAE 71

LIST OF FIGURES

Figure 1: Conceptual Framework: Intellectual Progression of Small-Retail Technology 9

Figure 2: System Architecture: Offline-First Synchronization Workflow 16

Figure 3: Figure 3: Existing Manual Sales and Inventory Management Workflow of Alekos' Corner General Merchandise 24

Figure 4: Proposed Sales and Inventory Management Workflow of Managing Sentry 25

Figure 5: Schematic diagram of conceptual framework of the Managing Sentry: A Point-Of-Sale and Inventory Management system for Alekos' Corner General Merchandise 27

Figure 6: Entity-Relationship Diagram of Managing Sentry 28

Figure 7: Iterative Development Process of Managing Sentry Using the Agile Methodology 29

LIST OF TABLES

Table 1: Comparative Analysis of Existing POS and Inventory Systems vs. Managing Sentry 22
Table 2: Software Development Tools 30
Table 3: Schedule and Timeline 34
Table 4: Responsibilities 35
Table 5:Budget and Cost Management 36
Table 5:Risk Management 37

CHAPTER 1
INTRODUCTION

1.1 Rationale of the Study

Retail businesses rely on efficient sales and inventory processes to maintain accurate records, serve customers promptly, and ensure that products remain available. As business operations become increasingly dependent on digital information, point-of-sale (POS) and inventory management systems have become important tools for organizing transactions, monitoring stock levels, and generating information that can support business decisions. The urgency of this study lies in the operational inefficiencies and financial risks associated with manual systems, including delayed inventory updates, misaligned stock levels, and revenue loss due to poor oversight. (Ramadhani et al., 2025) Delayed inventory updates can result in inaccurate information about the actual quantity of products available, while differences between recorded and actual stock levels can make it difficult for the business to determine when products need to be replenished. Poor monitoring of sales and inventory records may also make it difficult for the owner or managers to identify sales trends and possible revenue losses. These concerns are relevant to Alekos' Corner General Merchandise, where sales transactions, product pricing, inventory updates, and receipts are currently handled through manual and paper-based processes.

Despite the availability of digital technologies, some small businesses continue to depend on manual and paper-based processes for their daily operations. Inaccurate inventory amounts, waste from utilizing incorrect components in production, and the wrong pricing on all classes of inventory were the most frequent inventory errors accounted for by the participants.(Wynn, 2021) This issue is relevant to the development of

Managing Sentry, which is designed to support more organized inventory and business management. The application provides centralized records for products, inventory quantities, pricing, stock adjustments, and transactions. By maintaining updated inventory information and documenting stock changes, Managing Sentry aims to reduce discrepancies between recorded and actual inventory and provide more consistent information for managing products and their prices. The system's low-stock monitoring and stock adjustment features can also help users identify inventory concerns and maintain more reliable records.

These concerns are present at Alekos' Corner General Merchandise, a family-run general merchandise store operating under Heir Client Business Innovation. Based on the team's interview, direct observation, receipt and record review, and owner assessment, the store currently relies on paper receipts and manual procedures for recording sales and managing inventory. Transactions are recorded manually, while physical receipts are collected and stored by the owner. When errors occur, corrections may be made by crossing out information or writing directly on the receipt. Product pricing and inventory updates are also performed manually. According to the assessment conducted with the store owner, restocking and product pricing are among the most time-consuming tasks and are also prone to errors. Manual receipt writing can additionally slow down checkout operations, particularly during busy late-night transactions.

The current process affects several stakeholders within the business. The owner and the staff experience difficulty reviewing sales trends and maintaining physical records, while staff members spend additional time writing receipts, checking prices, and updating inventory manually. Customers may also experience longer waiting times when transactions are processed manually during busy periods. In addition, the lack of automated sales and

inventory reports limits the availability of timely information that management can use when monitoring business performance and planning restocking activities.

Unlike systems that focus only on processing sales transactions, Managing Sentry integrates point-of-sale operations, inventory monitoring, low-stock notifications, sales reporting, and offline transaction support into a single system designed around the operational needs of Alekos' Corner General Merchandise. This integration is intended to address the store's identified difficulties in manually recording sales, monitoring inventory, managing product prices, and accessing timely business reports. By connecting these functions, the system provides a more organized approach to recording transactions and monitoring the movement of products within the store. The proposed system is specifically designed to support the daily operations of the business while reducing its dependence on paper-based records and manual procedures. Therefore, this study aims to design and develop Managing Sentry, a customized Point-of-Sale and inventory management system that digitizes sales transactions, improves inventory monitoring, and supports more organized store operations at Alekos' Corner General Merchandise.

1.2 Statement of the Problem

Alekos' Corner General Merchandise currently relies on manual and paper-based processes for recording sales transactions, managing product prices, monitoring inventory, and storing receipts. These processes may result in slow checkout operations, errors in recording sales and pricing information, difficulties in maintaining accurate inventory records, and limited access to timely sales and inventory reports. The manual handling of receipts also makes it more difficult to organize and review transaction records, particularly during busy periods. In addition, the absence of automated inventory monitoring and reporting limits the ability of the owner and managers to efficiently monitor stock levels, identify products that require replenishment, and review sales performance. Therefore, there is a need for a system that can digitize sales transactions, improve inventory monitoring, and provide organized business records while supporting the operational needs of the store.

1.2.1 General Objective

The objective of this study is to design and develop Managing Sentry, a customized Point-of-Sale (POS) and inventory management system for Alekos' Corner General Merchandise that digitizes sales transactions, automates inventory monitoring, provides sales and inventory reports, and supports more organized store operations.

1.3 Significance of the Study

Owner and Managers. The owner and managers of Alekos' Corner General Merchandise may benefit from having easier access to organized sales and inventory information. Through the web-based administration portal, they can monitor product stocks, receive low-stock notifications, review sales trends, and manage product prices. Digital records can also reduce the dependence on physical receipts and make business information easier to

monitor and review.

Store Staff. Store staff may benefit from a more organized and efficient way of processing customer transactions. The Android-based POS application is designed to assist staff in selecting products, processing sales, and updating inventory without relying entirely on handwritten records. The system's offline functionality may also allow staff to continue recording transactions when an internet connection is unavailable.

Customers. Customers may benefit indirectly from faster and more consistent sales transactions. By reducing manual checkout processes and helping staff access updated product and pricing information, Managing Sentry may help minimize delays and reduce errors during transactions, particularly during busy periods.

The Client Business. Alekos' Corner General Merchandise may benefit from having a centralized system for sales and inventory management. The integration of POS, inventory monitoring, low-stock alerts, and sales reporting can help the business maintain more organized records and support better monitoring of its daily operations. The system is specifically designed around the client's identified needs and existing workflow.

Future Researchers and Developers. The study may serve as a reference for future researchers and developers who plan to develop similar POS, inventory management, or offline-capable business systems. The project demonstrates how a digital system can be designed around the operational requirements of a small, family-run retail business while integrating sales processing, inventory monitoring, and reporting features.

Scope and Limitations

This study focuses on the design and development of Managing Sentry, a Point of Sale (POS) and inventory management system intended for Alekos' Corner General Merchandise under Heir Client Business Innovation. The system covers the main sales and inventory activities of the business, including

transaction processing, product and price management, inventory monitoring, low-stock notifications, and sales reporting. It consists of an Android-based POS application for store staff and a web-based administration portal for the owner and managers. The system also supports offline transactions through a local database and synchronization when an internet connection becomes available.

The scope of the study is limited to the operational needs of a single retail business, specifically Alekos' Corner General Merchandise. The POS application is designed for manual product selection during checkout and does not include barcode scanners, weighing equipment, or other specialized hardware. The system is also intended for in-store transactions and focuses on cash-based or manually recorded sales. It does not integrate with external card terminals, online payment services, or banking APIs. Furthermore, the system does not include an automated discount application feature for senior citizens and persons with disabilities (PWDs). Any applicable discounts for these customers must be calculated and applied manually by the authorized staff during checkout, as the system is limited to recording transactions without automatically calculating or applying these specific discounts.

The study also has limitations that may affect the extent of the system's application. Since Managing Sentry is developed specifically according to the workflow and requirements of Alekos' Corner General Merchandise, its features and processes may not directly apply to businesses with different operational requirements. The system depends on manual product selection and data entry, which means that incorrect information entered by users may still result in inaccurate records. Although the system provides offline functionality and synchronization, some features that require updated online information may be limited when an internet connection is unavailable.

CHAPTER 2
RELATED SYSTEMS

This chapter reviews relevant studies, technical frameworks, and existing software related to Managing Sentry: A Point of Sale and Inventory Management System with Offline Synchronization for Alekos' Corner General Merchandise. In retail stores, depending on manual math, physical paperwork, and handwritten receipts often causes counter delays, pricing mistakes, and lost records. To solve these issues, this review looks into current research on Point of Sale (POS) systems, real-time stock tracking, and offline database synchronization. It also compares existing commercial, local, and open-source retail applications with the daily needs of small retail businesses. Through this comparison, the chapter identifies key gaps in current solutions, such as the lack of affordable systems that work without the internet while offering both simple mobile checkouts and web-based management tools. These findings explain why the proposed system is necessary and guide how it will be built.

2.1 Introduction and Theoretical Framework

In retail environments, Point of Sale and inventory control mechanisms serve as the operational backbone of business survival (Gauri et al., 2021). In micro, small, and medium enterprises, such as general merchandise stores and neighborhood outlets, retailers frequently depend on manual bookkeeping, pen-and-paper receipts, mental arithmetic, and physical ledger counts. As noted in empirical retail studies, relying on manual processes introduces recurring administrative bottlenecks, pricing inconsistencies, unrecorded inventory shrinkage, and vulnerability to catastrophic data loss due to misplaced or degraded paper records (Chanda & Mwanza, 2023; Wynn, 2021).

To address these vulnerabilities, software engineering has shifted toward digital Point of Sale systems with integrated inventory automation. However, conventional commercial solutions often assume reliable internet connectivity, dedicated proprietary POS hardware like barcode scanners and thermal printers, and recurring monthly licensing fees. These assumptions fail in developing retail ecosystems, particularly in provincial commercial hubs, where store owners encounter intermittent network infrastructure, tight operational margins, and space constraints at checkout counters.

Grounded in the Technology Acceptance Model and Distributed Database Synchronization Theory, this review traces the intellectual progression of retail automation, contrasts mobile application paradigms against legacy desktop platforms, evaluates offline-first local database architectures, and analyzes existing commercial, open-source, and localized Philippine retail applications. Through this inquiry, the chapter establishes the technical justification for the proposed system and identifies the operational gaps it is engineered to resolve.

2.2 Intellectual Progression and Thematic Literature Review

The development of modern retail systems spans several interrelated computing domains. The following thematic sections synthesize contemporary research across five major dimensions.

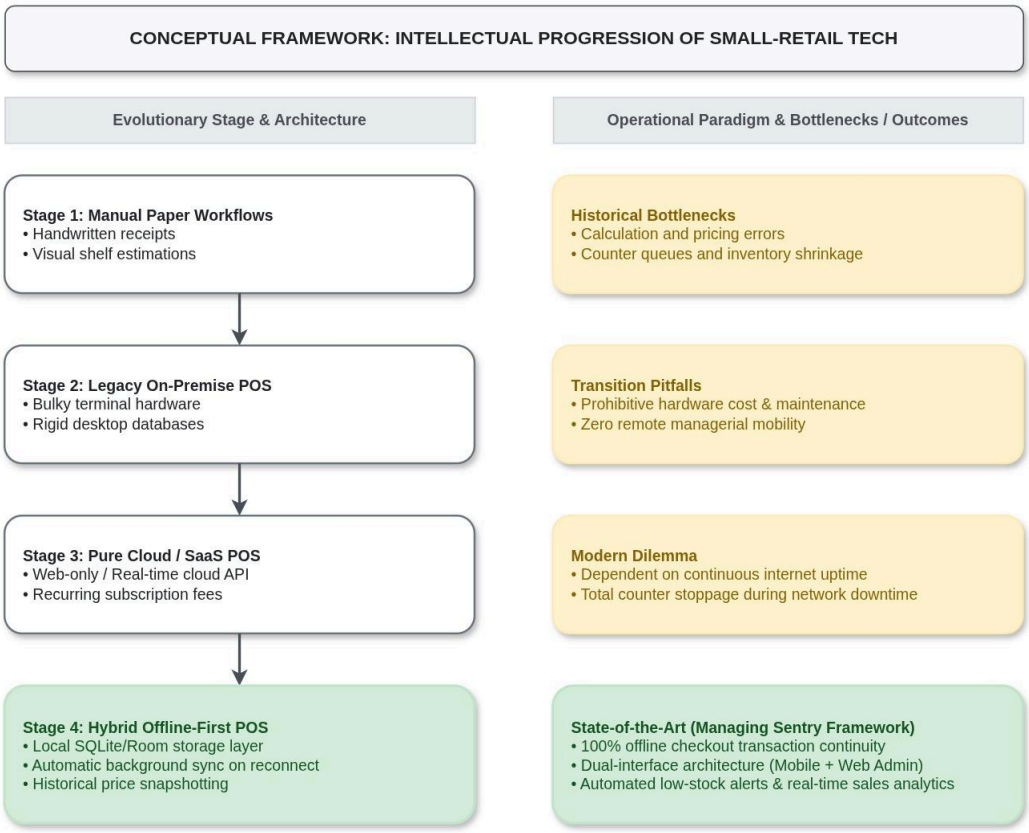


Figure 1: Conceptual Framework Illustrating the Evolution of Small-Retail Technology

2.2.1 Evolution from Manual Bookkeeping to Digital Point of Sale in Micro-Retail

Historically, small retail enterprises operated through traditional single-entry ledger systems. Transactions were recorded on physical paper books, and receipts were issued manually. Contemporary research highlights that manual bookkeeping in micro-retail creates major structural vulnerabilities:

1. Transaction Velocity Bottlenecks: Handwritten receipts slow down counter throughput during peak foot-traffic periods, creating long queues and customer dissatisfaction.
2. Computational and Human Error: Manual mental arithmetic and handwriting corrections produce compounding arithmetic discrepancies in gross sales and tax calculations (Wynn, 2021).
3. Physical Storage Degradation: Paper receipts are susceptible to water damage, fading ink, fire, misplacement, and intentional alteration.

The transition toward automated Point of Sale systems fundamentally transforms transaction processing from reactive bookkeeping to proactive data acquisition (Kholod et al., 2024; Yuniarto & Siregar, 2025). A Point of Sale system acts as the central node where retail transactions are finalized, inventory counts are decremented, and revenue balances are generated. Adopting an entry-level digital POS reduces checkout operational friction and eliminates basic calculation errors in small neighborhood retail stores (Ramadhani et al., 2025; Wahyuni, 2025). Furthermore, replacing physical paper receipts with electronic issuance via Android applications significantly reduces processing time and eliminates physical storage vulnerabilities (Gicale-Aguilar & Dasargo, 2022).

Point-of-sale efficiency is also important in improving the speed of retail transactions. Polasik et al. (2013) examined the time efficiency of different payment methods used at point-of-sale systems and found that payment methods differed in the time required to complete transactions, noting that contactless methods and mobile payments were competitive with cash. These findings demonstrate the importance of considering transaction efficiency when developing POS systems. In relation to Managing Sentry, the system is designed to support a more organized and efficient checkout process at Alekos' Corner General Merchandise by providing digital sales transactions, automated total calculation, and automatic inventory updates after completed sales. While Polasik et al. (2013) focused on the efficiency of specific payment methods, Managing Sentry broadens this concept by improving the overall sales workflow—automating the time-consuming manual calculations and integrating the transaction process directly with inventory management and sales records.

2.2.2 Inventory Management Models, Automated Restocking, and Discrepancy Control

Inventory represents the single largest liquid asset in any general merchandise retail enterprise. Inadequate inventory oversight results in two extremes: stockouts (leading to lost revenue and customer defection) and overstocking (leading to tied-up capital, expired goods, and storage degradation) (Boute et al., 2022).Modern inventory theory relies on algorithmic mechanisms such as Economic Order Quantity (EOQ), Reorder Point (ROP) thresholds, and safety stock buffers. Research demonstrates that implementing automated reorder alert levels within retail databases reduces stockout occurrences compared to stores using visual shelf inspections (Avlijas et al., 2021). When a

product's current balance $Q_{current}$ dips below its assigned safety threshold $R_{threshold}$, the system automatically flags the product for reordering:

$$R_{threshold} = (D \times L) + SS$$

Where:

D = Average daily demand rate

L = Supplier lead time in days

SS = Designated safety stock cushion

Furthermore, a critical challenge in retail inventory systems is inventory shrinkage—unrecorded stock loss resulting from shoplifting, product damage, expiration, or administrative omission. Traditional paper-based stores fail to detect shrinkage until end-of-year physical audits, by which time financial recovery is impossible (Chanda & Mwanza, 2023; Wynn, 2021). Integrating real-time stock-in/stock-out auditing and dynamic gross margin calculations allows retail management to trace product shrinkage to specific dates and operational shifts (Avlijas et al., 2021).

Small and medium enterprises (SMEs) may experience difficulties in managing inventory when they have limited digital infrastructure and continue to rely on manual inventory management methods. These difficulties can affect data accuracy, delay stock updates, and slow operational decision-making. (Shawn and Harefa ,2026) developed a web-based inventory management application for SMEs that aimed to improve operational efficiency through features such as stock visibility and accurate stock-in and stock-out transactions. The study

reported positive user satisfaction with the system's functional requirements and positive acceptance toward its adoption by SMEs. This study is relevant to Managing Sentry because Alekos' Corner General Merchandise similarly relies on manual processes for inventory and product management. Managing Sentry addresses these concerns by providing automated inventory monitoring, product and pricing management, stock adjustments, and low-stock notifications. Similar to the study of (Shawn and Harefa, 2026), Managing Sentry aims to improve the accuracy and visibility of inventory information; however, it is specifically designed around the operational requirements of Alekos' Corner General Merchandise and integrates inventory management with point-of-sale transactions, sales reporting, and offline transaction support.

Similarly, manual inventory management can create difficulties in maintaining accurate and timely stock information. Wuryanto et al. (2026) reported that PT Interior Indonesia's reliance on manual stock recording resulted in data inaccuracies, delays in information, and difficulties in monitoring stock availability in real time. To address these concerns, the researchers developed an Android-based inventory application that improved the accuracy of stock recording, accelerated stock management processes, and supported real-time monitoring of stock availability. This study demonstrates how a digital inventory system can help address problems associated with manual stock management. Managing Sentry is designed to address the manual inventory and sales processes of Alekos' Corner General Merchandise by providing automated inventory monitoring, product and pricing management, stock adjustments, and low-stock notifications. However, unlike the strictly inventory-focused application described by Wuryanto et al. (2026), Managing Sentry integrates inventory management with point-of-sale transactions, sales reporting, and offline transaction support to comprehensively address the specific operational requirements of the client business.

2.2.3 Mobile-First POS Checkout vs. Comprehensive Web Admin Portals

In retail systems design, role-based functional separation is vital for operational security and efficiency. Mobile applications tailored for frontline cashiers—such as Android-based POS solutions explicitly designed for micro-enterprises—must be lightweight, focused, and optimized for high-speed transaction processing (Karandika, 2026). Conversely, administrative controls require expansive web interfaces to centralize cashier management, inventory oversight, and catalog data (Choudhury, 2022; Umam, 2025).

For Managing Sentry, this architectural division is strictly implemented:

- Android POS Application (Staff-Facing): Built exclusively for frontline staff to execute counter transactions. It provides three core operational views: (1) The POS Dashboard / Cart Checkout, where clerks select items, compute totals, accept cash tenders, and calculate customer change; (2) Product View-Only List, allowing staff to check catalog items and prices without modifying catalog master data; and (3) Transaction History, enabling clerks to review past shift receipts. Crucially, the mobile app operates 100% offline using a local SQLite/Room database.
- Web Admin Portal (Management-Facing): Built for the store owner and administrators via desktop or browser. The web portal houses full administrative control: creating new products, configuring pricing markups, updating inventory stocks, logging restock shipments, tracking capital balances, and generating comprehensive daily/weekly sales analytics and business reports.

2.2.4 Offline-First Database Paradigms and Data Synchronization Architectures

A major debate in contemporary software engineering centers on Pure Cloud Architecture versus Offline-First Hybrid Architecture. Cloud-native SaaS applications offer seamless cross-device synchronization and centralized data persistence. However, pure cloud systems exhibit a single catastrophic point of failure: network dependency (Pothineni, 2024).

In developing markets and local store environments, internet connections are frequently erratic. If a POS terminal requires an active HTTP/REST API connection to finalize a sale, network downtime freezes retail operations, leaving staff unable to process checkouts or compute customer change.

To solve this dilemma, software engineers advocate for the Offline-First Paradigm (Pothineni, 2024). In Managing Sentry, the Android POS application treats the local on-device SQLite database (managed via Android Room) as the primary source of truth for all immediate read/write operations during store hours. Every transaction, cart checkout, and stock decrement is committed instantly to local storage, guaranteeing sub-second response times and 100% operational availability regardless of internet status.

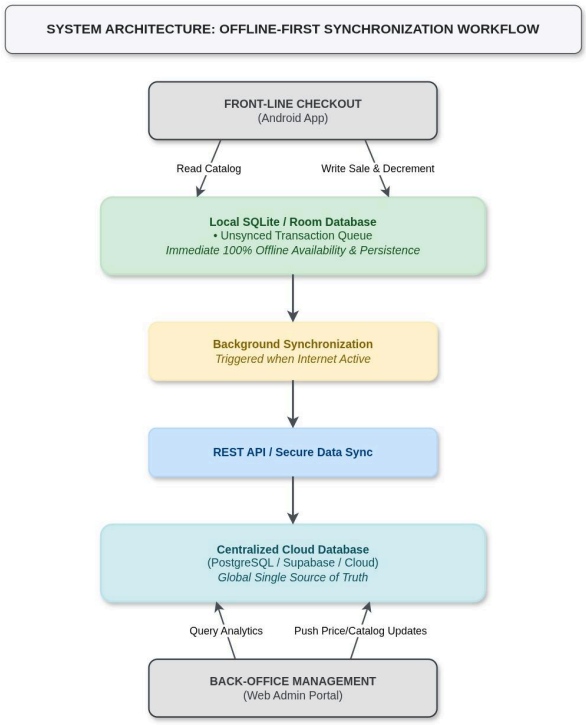


Figure 2: System Architecture Illustrating the Offline-First Synchronization Workflow

Synchronization occurs asynchronously via background workers whenever network connectivity is restored. Conflict resolution models ensure mathematical consistency: transactional "deltas" (e.g., subtracting sold quantities from inventory counts) are queued and synced to the centralized cloud database, preventing lost updates during multi-device operations (Pothineni, 2024; Uriawan, 2024).

2.2.5 Analytical Dashboards, Sales Forecasting, and Operational Decision Support

Beyond basic transaction processing, modern POS frameworks function as strategic management information systems. Unrecorded or siloed transaction data represents wasted business intelligence. When sales are digitized through a centralized web portal, transaction records generate actionable business intelligence through automated dashboard analytics (Kholod et al., 2024; Maharani, 2026).

Key analytical metrics highlighted in modern retail management include:

- Fast-Moving vs. Slow-Moving SKU Classification: Identifying high-velocity products that drive daily cash flow versus dead capital trapped in stagnant inventory.
- Hourly Sales Velocity: Pinpointing exact rush hours (e.g., peak late-night sales) to optimize staffing schedules and counter preparation.
- Dynamic Cash Balance Auditing: Integrated computerized systems with stock monitoring and income reporting capabilities allow retail management to dynamically audit expected drawer balances and reconcile shift revenues, eliminating discrepancies caused by unrecorded personal withdrawals or untracked cash payouts (Chanda & Mwanza, 2023; Nelvic, 2026).

2.3 Review of Related Systems

To contextualize the development and feature set of Managing Sentry, this section evaluates existing commercial systems, open-source enterprise solutions, and localized retail applications.

2.3.1 Square POS

Square POS (Block, Inc., USA) is one of the most widely adopted point-of-sale platforms globally. It provides a mobile and tablet application paired with cloud management dashboards.

- Core Features: Digital cash register, catalog and category organization, cloud sales reporting, automated tax calculation, and payment processing.
- Architectural & Operational Strengths: Square delivers a polished user interface, rock-solid cloud data resilience, and rich sales analytics. It supports an "Offline Mode" that permits credit card transactions and sales to be queued locally during temporary outages.
- Limitations in Small Local General Merchandise: Square is tightly coupled with proprietary payment hardware and payment processing fees. Furthermore, its offline mode was architected for intermittent drops in modern urban broadband rather than prolonged offline usage; transactions must be uploaded within a strict 24-hour window or they expire. Its feature set is also overly complex for family-run retail stores in developing countries that deal strictly in cash transactions and do not use payment processing terminals.

2.3.2 Loyverse POS

Loyverse POS is an international cloud-based mobile point-of-sale application designed for small cafes, boutiques, and retail shops.

- Core Features: Smartphone/tablet POS app, web-based Back-Office dashboard, stock tracking, multi-store support, and employee time tracking.
- Architectural & Operational Strengths: Loyverse features a dual architecture consisting of an Android/iOS mobile client and a web-based administrative portal. It supports basic offline sales recording and queues transactions locally on the device.

- Limitations in Small Local General Merchandise: Although the basic checkout app is free, core operational modules—including advanced inventory management, historical purchase order audits, and employee shift permission control—require expensive monthly recurring add-on subscriptions. For micro-merchants operating on slim retail margins, ongoing subscription fees represent an unsustainable overhead cost. Additionally, syncing multiple offline devices without active internet often causes inventory calculation desynchronization.

2.3.3 Odoo POS (Community Edition)

Odoo is an open-source, modular Enterprise Resource Planning (ERP) platform developed in Python and PostgreSQL, featuring an integrated web-based Point of Sale module.

- Core Features: Real-time integration with double-entry inventory accounting, automated stock procurement rules, warehouse tracking, and multi-currency pricing.
- Architectural & Operational Strengths: Odoo POS runs directly inside modern web browsers using web cache and HTML5 local storage, permitting checkout operations to proceed when the browser loses network access. Once reconnected, the browser syncs sales orders back to the core ERP backend.
- Limitations in Small Local General Merchandise: Odoo's technical footprint is substantial. Deploying and maintaining an on-premise or cloud Odoo server requires dedicated Linux server administration, extensive database configuration, and substantial hardware resources (minimum 4GB–8GB RAM server hosting). The user interface is dense, overwhelming non-technical store clerks with enterprise accounting jargon, making it impractical for small single-branch general merchandise stores.

2.3.4 Peddlr

Peddlr is a Philippine-developed mobile POS and micro-accounting application designed specifically for Filipino micro-enterprises, including sari-sari stores, wet market stalls, and small retail outlets.

- Core Features: Offline mobile POS transactions, simple ledger management, credit (*utang*) tracking, inventory alerts, and expense logging.
- Architectural & Operational Strengths: Peddlr was built explicitly to address the realities of Philippine micro-retail. It runs natively on standard budget Android smartphones, works without an active mobile data connection, and replaces physical paper credit notebooks.
- Limitations in Small Local General Merchandise: Peddlr is strictly mobile-centric; it lacks a robust, dedicated desktop/web administrative portal where a store owner can manage catalog data, adjust bulk pricing markups, analyze graphical sales trends, and configure reorder points on a larger computer screen. Managing large inventory catalogs on a 6-inch phone screen creates ergonomic friction for owners. Furthermore, transaction data is stored inside a closed ecosystem, offering limited data portability and customization for stores with specialized multi-role staff workflows.

2.3.5 Utak POS

Utak POS is a Philippine commercial cloud POS and back-office inventory system widely utilized by local food kiosks, retail stores, and service shops.

- Core Features: Tablet-based point of sale, real-time inventory tracking, staff attendance monitoring, gross sales reporting, and back-office web reporting.
- Architectural & Operational Strengths: Utak POS provides dedicated customer support within the Philippines, handles local tax formatting, and

offers a clean web-based manager portal for real-time sales visibility across devices.

- Limitations in Small Local General Merchandise: Utak POS operates on a commercial subscription model (typically charging monthly fees per terminal), which is cost-prohibitive for small family-run retail stores. Crucially, Utak POS is primarily cloud-reliant; unexpected internet service outages in local neighborhoods instantly disrupt terminal availability and halt counter sales.

2.4 Comparative Analysis of Related Systems

To establish the technological positioning of Managing Sentry, Table 1 evaluates the features, technical limitations, and architectural differences across existing systems and the proposed system.

Table 1. Comparative Analysis of Existing POS and Inventory Systems vs. Managing Sentry

System Name	Platform Support	Offline Operation Capability	Inventory & Low-Stock Alerts	Web Admin & Reporting Portal	Pricing Model / Licensing
Square POS	Mobile (iOS/Android) & Web	Limited (24-hour max offline queue)	Basic stock tracking	Cloud Web Dashboard	Free app + High payment processing fees
Loyverse POS	Mobile (iOS/Android) & Web	Basic offline local queue	Paid add-on module	Web Back-Office	Freemium with expensive monthly add-ons
Odoo POS	Web Browser / Enterprise Server	HTML5 browser cache storage	Fully integrated double-entry ERP	Comprehensive Web ERP	Open-source (Heavy server hosting costs)
Peddlr	Mobile Only (Android/iOS)	Full offline mobile ledger	Basic inventory count	Limited mobile reporting	Free / Freemium mobile app
Utak POS	Tablet & Web Cloud	Cloud-reliant (Fails during outages)	Real-time tracking	Web Management Portal	Commercial monthly subscription per terminal
Managing Sentry (Proposed)	Android Mobile App + Web Admin	100% Offline-First (SQLite Room + Background Sync)	Automated low-stock alerts with reorder estimates	Dedicated Web Portal with live sales trends & logs	Custom Academic Solution (Tailored for Alekos' Corner)

2.5 Synthesis of Reviewed Literature and Systems

The synthesis of contemporary literature and existing commercial software reveals clear patterns and critical gaps in retail technology:

1. Peddlr handles offline mobile transactions well, but lacks comprehensive web management interfaces. Managing Sentry bridges this gap by providing frontline staff with an offline-first Android POS app (featuring cart checkout, product viewing, and transaction history) backed by automatic background synchronization (Pothineni, 2024).
2. Dual-Interface Functional Separation: Existing systems are often polarized: either mobile-only apps that constrain managerial oversight on small screens, or heavy desktop/cloud ERP systems (such as Odoo) that overwhelm non-technical users. Managing Sentry implements a clean functional division: store staff utilize the mobile app exclusively for rapid offline checkouts, product catalog viewing, and history checks, while store owners utilize the dedicated Web Admin Portal for full product creation, pricing updates, restock logging, financial capital tracking, and advanced sales analytics.
3. Elimination of Paper Vulnerability: As highlighted by historical and current research, manual bookkeeping results in calculation errors and lost paper receipts (Wynn, 2021). Managing Sentry replaces physical paper records with secure digital logs and historical price snapshots, ensuring absolute data integrity, accurate change computation, and transparent financial auditing for Alekos' Corner General Merchandise.

CHAPTER 3

DESIGN AND METHODOLOGY

This chapter presents the design and methodology used in the development of Managing Sentry, a customized Point-of-Sale (POS) and inventory management system for Alekos' Corner General Merchandise. It describes the conceptual framework, system analysis and design, development model, development approach, and software development tools used in developing the system.

3.1 Conceptual Framework

The conceptual framework of Managing Sentry describes the transition from the existing manual business process to a computerized POS and inventory management process. The framework identifies the current procedures followed by Alekos' Corner General Merchandise and the proposed procedures that will be supported by the system. The current process at Alekos' Corner General Merchandise is primarily manual. When a customer purchases a product, the staff writes the transaction on a receipt and manually calculates the total amount. The owner keeps the physical receipt for record keeping. After sales transactions, the staff manually checks the available inventory and determines when products need to be restocked.

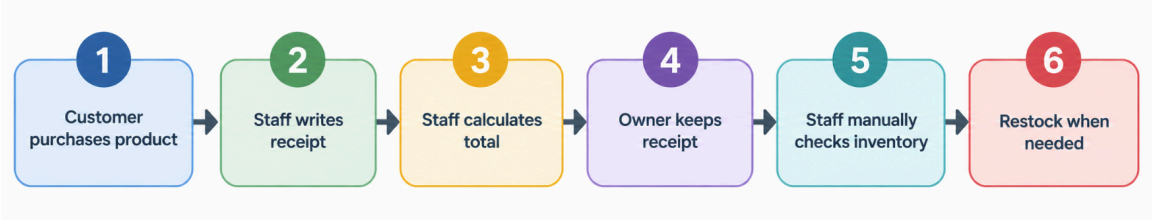


Figure 3: Existing Manual Sales and Inventory Management Workflow of Alekos' Corner General Merchandise

This process requires manual recording and checking of sales and inventory information, which can make monitoring and record management more dependent on physical documents and manual procedures.

Managing Sentry is designed to replace the manual sales and inventory procedures with a computerized process. The staff will use the Android application to process customer purchases. The system records the transaction and automatically updates the corresponding inventory quantity. The system also monitors inventory levels and product expiration dates. When the remaining quantity of a product reaches 10% or less of the originally inputted stock quantity, the system automatically detects the product as low stock. A red blinking button is displayed on the Super Admin side to indicate the low-stock condition.The system also monitors product expiration dates. When a product is within one week of its expiration date, the system provides an expiration alert so that the Super Admin/Owner can identify products that are approaching expiration.When products are restocked, the new stock quantity can be recorded in the system. The restocking activity is retained in the inventory history, where notes can be added to document information about the stock adjustment.

The proposed process can therefore be represented as:

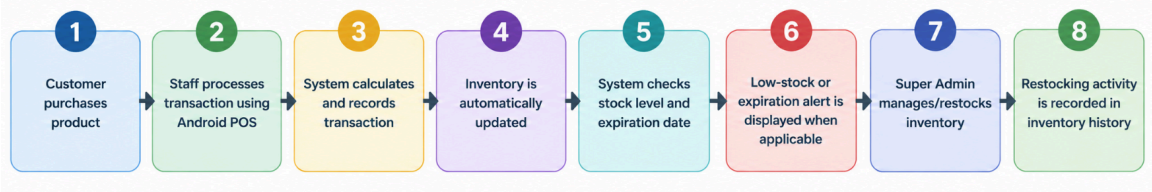


Figure 4: Proposed Sales and Inventory Management Workflow of Managing Sentry

Managing Sentry also provides the Super Admin/Owner with sales information that can be used for business monitoring. The reporting functionality includes:

1. Daily sales
2. Weekly sales
3. Monthly sales
4. Total sales revenue
5. Products sold
6. Best-selling products

These reports provide organized sales information that can be accessed digitally instead of relying solely on manually stored receipts.

Managing Sentry is also designed to support sales transactions when an internet connection is unavailable. Staff can continue processing transactions through the Android application. The transaction and related inventory changes are temporarily handled on the device and are synchronized with the main database once internet connectivity becomes available.

Note: The specific local-storage technology for offline transactions has **NOT YET BEEN DECIDED** and should be specified once the development team selects the technology.

Conceptual Framework Diagram

The conceptual framework diagram should show the relationship between the proposed Managing Sentry process, and the resulting improvements in sales and inventory management.



Figure 5: Schematic diagram of conceptual framework of the Managing Sentry: A Point-Of-Sale and Inventory Management system for Alekos' Corner General Merchandise

3.2 Analysis and Design

Managing Sentry follows an analysis and design process that identifies the required system functions, users, data relationships, and system components before and during development. The system is designed around the operational requirements of Alekos' Corner General Merchandise and consists of an Android application for staff and a web-based Super Admin portal.

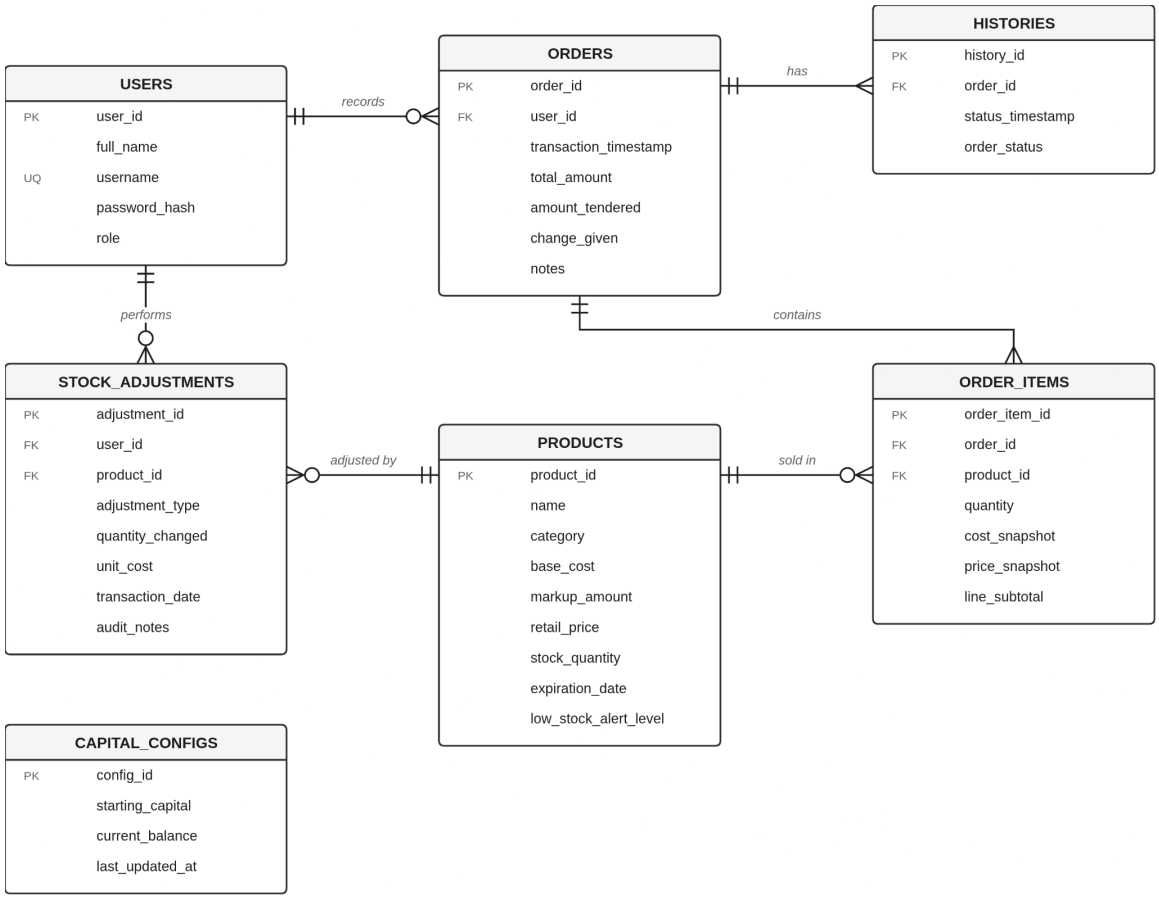


Figure 6: Entity-Relationship Diagram of Managing Sentry

3.3 Development Model

The development process follows an iterative cycle:

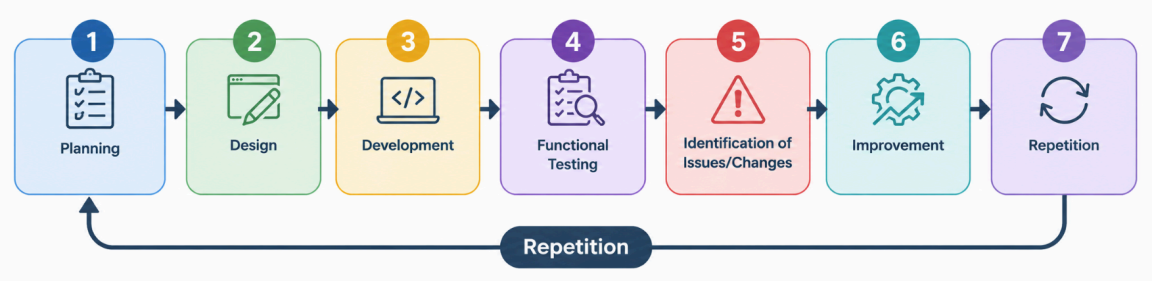


Figure 7: Iterative Development Process of Managing Sentry Using the Agile Methodology

Under this approach, the development team works on system functions and continuously evaluates the results. After a feature is developed, functional testing is performed to determine whether the feature works according to its intended function. Issues or required changes are then identified and addressed before continuing with further development. The Agile approach is applied to the different functional areas of Managing Sentry, including the POS, inventory management, alerts, sales reporting, and offline transaction functionality.

3.4 Development Approaches

Managing Sentry uses a **Top-down Development Approach**. The Top-down approach begins with the overall system and divides it into smaller components or functional areas. For Managing Sentry, the overall objective of developing a customized POS and inventory management system is divided into specific system functions.

3.5 Software Development Tools

Software/Technology	Version	Purpose	Reason for Selection
Android Studio	2026.1.4 (Quail 4)	Used to develop the Android Point-of-Sale (POS) application.	Provides tools for Android application development, debugging, and testing.
Java	Latest stable JDK version (21 or higher)	Used to implement the logic and functionality of the Android application.	The team has prior knowledge of Java from coursework, making it suitable for Android development.
XML	Android-supported XML	Used to design the layouts and user interface of the Android application.	Separates interface layouts from application logic and supports Android UI development.
HTML	HTML5	Used to structure the web-based administration portal.	Provides the standard structure for web pages and administrative interfaces.

CSS	CSS3	Used to style the web-based administration portal.	Allows the team to create a consistent, organized, and user-friendly interface.
JavaScript	Latest stable ECMAScript standard (ES6 or higher)	Used to implement interactive features on the web-based administration portal.	Supports dynamic content and user interaction.
PHP	Latest stable PHP 8.x	Used for server-side processing and database operations, where applicable.	Supports backend development and communication with the database.
FastAPI	fastapi[standard]==0.112.0	Used to develop API endpoints for communication between system components.	Simplifies API development and supports data validation.
Retrofit	Retrofit (Version)**:** `3.0.0`	Used by the Android application to communicate with API endpoints.	Simplifies sending requests to APIs and receiving responses from the backend.

PostgreSQL	PostgreSQL (Version): 18.6	Used to store product information, sales transactions, inventory records, user accounts, and other system data.	Provides a relational database for organizing and maintaining business records.
Supabase	Latest platform version (managed by Supabase)	Used to provide the hosted PostgreSQL database and related backend services.	Offers a managed database environment that reduces the need to maintain a separate database server.

Android Development

Android Studio is used as the development environment for the Android application. The Staff POS application is developed using **Java**, while **XML** is used to design the Android user interface.

Android Studio provides the development environment needed to build and test the Android application.

Web Development

The Super Admin web portal uses:

- **HTML** for structuring web content
- **CSS** for presentation and styling
- **JavaScript** for client-side functionality

These technologies are used to develop the web-based interface used by the Super Admin/Owner.

Backend and API

The system uses **PHP and FastAPI** for backend/API-related functionality. These technologies provide the communication layer between the client applications and the database. The Android application communicates with the backend through **Retrofit**, which is used to make API requests from the Android application.

Retrofit

Retrofit is used in the Android application for API communication. It allows the Android Staff application to communicate with the backend services responsible for processing requests and interacting with the system's database.

Database

Managing Sentry uses **PostgreSQL** as its relational database management system. The PostgreSQL database is hosted through **Supabase**, which provides the hosted database environment used by the system. The database is responsible for storing the system's structured information and supporting the data requirements of the POS, inventory, user, and reporting functions.

System Architecture

The major components of Managing Sentry communicate through the backend/API layer. The Android Staff application follows this structure:

Android Staff App → Retrofit → PHP/FastAPI API → Supabase/PostgreSQL

The Super Admin web portal follows this structure:

Super Admin Web Portal → PHP/FastAPI API → Supabase/PostgreSQL

This architecture allows the client applications to communicate with the database through the backend/API layer rather than directly accessing the database.

3.6 Project Management

This section presents how the development team organizes and manages the activities required to complete Managing Sentry. It covers the project schedule and timeline, the responsibilities assigned to team members, and the management of the project's budget and costs. These areas help the team monitor the progress of development, distribute tasks appropriately, and ensure that available resources are used efficiently.

3.6.1 Schedule and Timeline

Development was carried out across structured phases, beginning August 3, 2026, each tied to specific tasks and deliverables, consistent with the sprint-based planning approach used throughout the project.

Phase / Task	Timeline
Requirements, Client Validation, Project Planning & Documentation	1st week of August 2026
UI/UX Design, Backend/API Development, Android POS Development	2nd–4th week of August 2026
Database Design Setup & Web Admin Portal Development	1st–2nd week of September 2026

Inventory Management Module Development	1st–2nd week of September 2026
Offline Database, Synchronization, Reporting & Sales Analytics	2nd–4th week of September 2026
Integration Testing	2nd–4th week of September 2026
Bug Fixing & Refinement	2nd–4th week of September 2026
Client Testing / Acceptance	1st–2nd week of October 2026
Final Documentation, Presentation & Deployment	1st–2nd week of October 2026

Progress across these phases was supported by weekly team stand-ups and bi-weekly client feedback sessions, allowing the team to validate work against client expectations before advancing to the next phase.

3.6.2 Responsibilities

Project responsibilities were distributed across the team according to each member's primary technical and documentation role, as defined in the project charter:

Team Members	Role	Key Responsibilities
Aldave, Resty S.	Design Lead / Frontend Developer / QA / Technical Lead	UI/UX design, mobile interface, system testing, application logic, API integration
Llaguno, Clary Mae F.	Database Lead / Database Developer	Table schemas, query optimization, storage & backups
Rama, John Wilfred Y.	Project Manager / Documentation Lead /	Timelines, client communications, project

	Technical Lead	documentation, application logic, API integration,
Bagahansol, Shan Lester S.	Documentation	Documentation

3.6.3 Budget and Cost Management

A total budget allocation of ₱25,000.00 was set for the project, covering infrastructure, tools, and documentation costs incurred during development:

Category	**Allocation**
Cloud Database & Web Hosting	₱8,500.00
Testing Devices & UI Tools	₱5,500.00
Documentation & Site Visits	₱4,000.00
Contingency Reserve (~28%)	₱7,000.00
Total	**₱25,000.00**

The client, Jose Mari A. Matas (Alekos' Corner General Merchandise), was responsible for requirements validation, scope approval, and acceptance testing, while academic supervision and project oversight were provided by the course instructor, Jana Lency Q. Pino. Involving the client directly in defining project scope and success criteria reflects the broader finding that engaging

3.6.4 Change and Risk Management

The project management plan includes procedures for handling additional changes requested by the client and addressing potential risks that may arise during the development and implementation of Managing Sentry. Any requested changes will be documented, evaluated based on their feasibility, compatibility

with the existing system, development time, and project scope, and discussed with the client before implementation. One possible enhancement is the addition of a customer receipt issuance feature. If the business obtains the necessary permits or authorization and satisfies the applicable legal and tax requirements, the development team may modify the system to generate and print customer receipts for completed transactions. This enhancement will be evaluated based on the client's needs, available resources, and compatibility with the Android POS application. Before deployment, the feature will undergo functional testing to verify the accuracy of transaction details, receipt formatting, and printing operations.

Risk management identifies possible events that could negatively affect the development or operation of Managing Sentry. The team will monitor potential risks throughout development and prepare appropriate responses to minimize their impact.

Potential Risk	Possible Effect	Planned Response
Internet connection failure	Online communication or synchronization may be interrupted	Support offline transaction processing and synchronize records when connectivity is restored
Development errors	Features may not work as intended	Perform functional testing and correct identified issues
Database problems	Data may become unavailable or synchronization may be affected	Monitor database connectivity and verify database operations during testing
Client-requested changes	Changes may affect the schedule or existing functionality	Evaluate and document changes before implementation
Limited development	Some features may not	Prioritize required

time	be completed on schedule	features and monitor project progress
Incorrect inventory data	Inventory information may become inaccurate	Test inventory updates and verify stock adjustments
Missed expiration or low-stock conditions	Products may not be identified when action is needed	Test the alert conditions using appropriate inventory and expiration data
Loss of internet during transactions	Transactions may not immediately reach the main database	Allow transactions to be handled offline and synchronize them once connectivity returns
Lack of necessary permits or failure to meet applicable receipt issuance requirements	The business may be unable to use the system to issue official receipts legally.	Verify applicable legal, tax, and business requirements with the relevant authorities before implementing the receipt issuance feature.
Incorrect transaction details on receipts	Customers may receive inaccurate information regarding purchased products, quantities, prices, or totals.	Conduct functional testing to verify receipt accuracy and ensure that printed details match the recorded transaction.
Receipt printer malfunction or incompatibility	Receipts cannot be printed, potentially delaying checkout operations.	Test printer compatibility before deployment, maintain the printing device, and provide a way to retrieve or reprint receipts when appropriate.
Additional receipt printing costs	Increased operating expenses for printer equipment, paper, ink, and maintenance.	Inform the client of the expected expenses and obtain approval before purchasing equipment or implementing the feature.

Receipt printing failure during a transaction	Customers may not receive a printed receipt even though the sale has been recorded.	Preserve the transaction record in the system and allow the authorized staff to retrieve or reprint the receipt when printing becomes available.

Risk monitoring will continue throughout the development process. When a risk occurs, the team will assess its effect and apply the appropriate response while documenting significant issues and changes.

3.7 Verification, Validation and Testing

3.7.1 Verification

Verification determines whether the development team is developing the system right. The team will compare the implemented system with the documented requirements, system design, database structure, and intended functions of Managing Sentry.

The planned verification activities include:

1. **Requirements Verification**
   Review the documented system requirements and compare them with the functions being developed to ensure that the required features are being implemented.
2. **Database Verification**
   Compare the implemented PostgreSQL database with the approved Entity-Relationship Diagram (ERD) and relational schema to verify that the required entities, attributes, primary keys, foreign keys, and relationships are properly implemented.

3. **Transaction Verification**

   Verify that sales transactions entered through the Android POS are properly recorded and that the corresponding inventory quantities are updated.

4. **Low-Stock Verification**

   Verify that the system detects a product when its remaining stock reaches **10% or less of the originally inputted stock quantity** and that the corresponding red blinking indicator appears on the Super Admin side.

5. **Expiration Alert Verification**

   Verify that the system identifies products that are within **one week of their expiration date** and displays the appropriate alert.

6. **Restocking Verification**

   Verify that restocked quantities are correctly added to inventory and that the restocking activity is recorded in the inventory history with the applicable notes.

7. **Reporting Verification**

   Verify that the sales reports correctly reflect recorded transactions, including daily, weekly, and monthly sales, total sales revenue, products sold, and best-selling products.

8. **API and System Communication Verification**

   Verify that the Android application and Super Admin web portal communicate correctly with the PHP/FastAPI backend and PostgreSQL database through the defined system architecture.

9. **Offline Synchronization Verification**

   Verify that transactions and related inventory changes processed while offline are synchronized with the main database after internet connectivity is restored.

These activities allow the team to determine whether the system has been implemented according to its specified requirements and design.

3.7.2 Validation

Validation determines whether the team is **developing the right system** for the intended users and business requirements of Alekos' Corner General Merchandise. The validation activities will focus on determining whether Managing Sentry addresses the problems identified in the existing manual process and whether its functions are useful to the Staff and Super Admin/Owner.

The planned validation activities include:

1. **Client Requirements Review**
   Compare the completed system functions with the requirements identified from Alekos' Corner General Merchandise.
2. **Process Validation**
   Determine whether the system supports the actual business process, from processing a customer's purchase through recording the transaction and updating inventory.
3. **Staff Validation**
   Evaluate whether the Android POS supports the Staff's intended responsibilities for processing customer transactions.
4. **Super Admin Validation**
   Evaluate whether the web portal provides the Super Admin/Owner with the required inventory monitoring, alerts, restocking history, and sales reports.
5. **Inventory Monitoring Validation**
   Determine whether the low-stock and expiration monitoring functions provide information that is useful for managing products.

6. **Reporting Validation**
   Determine whether the available reports provide the sales information needed by the business, including daily, weekly, and monthly sales, total sales revenue, products sold, and best-selling products.
7. **Client Feedback**
   Gather feedback from the intended client regarding whether the system functions address the problems encountered in the existing manual process.
8. **Improvement Validation**
   Review feedback and identified concerns and determine whether changes or improvements are required before the system is considered ready.

Validation therefore focuses on the suitability and usefulness of the system rather than only checking whether the software technically works.

3.7.3 Functional Testing

Functional testing will be used to determine whether each feature of Managing Sentry performs according to its intended function. The testing process will be performed throughout development so that errors can be identified and corrected as part of the Agile development cycle.

The planned functional tests include:

1. **Sales Transaction** – Process a customer purchase through the Android POS. The expected result is that the transaction is correctly recorded.
2. **Total Calculation** – Add selected products to a transaction. The expected result is that the system calculates the correct transaction total.
3. **Inventory Update** – Complete a product sale. The expected result is that the corresponding inventory quantity decreases correctly.

4. **Low-Stock Detection** – Reduce inventory to 10% or less of the original quantity. The expected result is that the system identifies the product as low stock.
5. **Low-Stock Indicator** – Trigger the low-stock condition. The expected result is that a red blinking indicator appears on the Super Admin side.
6. **Expiration Monitoring** – Test a product that is within one week of expiration. The expected result is that the system displays an expiration alert.
7. **Restocking** – Add stock to an existing product. The expected result is that the inventory quantity is updated correctly.
8. **Stock History** – Record a restocking activity with notes. The expected result is that the restocking activity appears in the inventory history.
9. **Sales Reporting** – Generate daily, weekly, and monthly reports. The expected result is that the correct sales information is displayed.
10. **Revenue Reporting** – Review recorded sales revenue. The expected result is that the total sales revenue corresponds to the recorded transactions.
11. **Products Sold** – Review product sales information. The expected result is that the products sold are correctly represented.
12. **Best-Selling Products** – Generate best-selling product information. The expected result is that products are ranked according to their recorded sales.
13. **Offline Transaction** – Process a transaction without internet connectivity. The expected result is that the transaction can be handled offline.
14. **Synchronization** – Restore the internet connection after an offline transaction. The expected result is that the transaction and related inventory changes synchronize with the main database.

Errors discovered during functional testing will be documented and addressed by the development team. After corrections are implemented, the

affected functionality will be tested again to determine whether the issue has been resolved.

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APPENDICES

APPENDIX A

PROJECT CHARTER

Project Title: **Managing Sentry**
Project Start Date:**August 3, 2026**
Key Schedule Milestones: Requirements, Planning & Design:1st – 4th week of August 2026 Development (POS, Backend, Offline Sync & Analytics): 2nd week of August – 4th week of September 2026 Testing & Bug Fixing (Integration & Client Acceptance): 2nd week of September – 2nd week of October 2026 Deployment & Final Presentation:1st – 2nd week of October 2026
Budget Information: Total Allocation: ₱25,000.00 Cloud Database & Web Hosting: ₱8,500.00 Testing Devices & UI Tools: ₱5,500.00 Documentation & Site Visits: ₱4,000.00 Contingency Reserve (~28%): ₱7,000.00
Project Manager: Rama, John Wilfred Yu

Project Objectives:

Digitize Checkout: Replace manual paper receipts with an offline-capable Android POS application.

Automate Inventory: Implement real-time stock tracking, low-stock alerts, and reorder estimates.

Centralize Management: Build a Web Admin Portal for live sales analytics and trend reporting.

Secure Data: Protect transaction records using local storage backup and cloud synchronization.

Main Project Success Criteria:

100% digital counter transactions completed without paper receipts.

Automatic real-time stock updates and functional low-stock notifications.

Seamless offline operation with automatic data synchronization upon internet reconnect.

Accurate daily/weekly sales reports generated via the Web Admin Portal.

Approach:

Develop a tailored, offline-first hybrid system (Android POS + Web Admin Portal) using a Supabase database.

Execute sprint deliverables with weekly team stand-ups, bi-weekly client feedback sessions, and pre-deployment QA testing.

Roles & Responsibilities:

Rama, John Wilfred Yu –
| Project Manager / Documentation Lead | Timelines, client communications, project documentation | Technical Lead / Backend Developer | Application logic, API integration, final presentation |

Aldave, Resty S. –
| Design Lead / Frontend Developer / QA | UI/UX design, mobile interface, system testing | Technical Lead / Backend Developer | Application logic, API integration, final presentation |

Llaguno, Clary Mae F. –
| Database Lead / Database Developer | Table schemas, query optimization, storage & backups |

Jose Mari A. Matas –
| Client / Store Owner | Requirements validation, scope approval, acceptance testing |

Jana Lency Q. Pino –
| Course Instructor | Academic supervision and project oversight |

Comments:

Course & Section: BSIT2-B1 (Client-Based Academic Project)

Client Organization: Heir Client Business Innovation (Alekos' Corner General Merchandise)

Submission Date: September 5, 2026

APPENDIX B

TRANSMITTAL LETTER

October 08, 2026

Jana Lency Q. Pino

Faculty Adviser
Southwestern University PHINMA

Dear Ms Pino,

We, the second-year students of the College of Information Technology under the supervision of Sir Jimuel, respectfully submit our IT project proposal entitled **"Managing Sentry: A Point-of-Sale and Inventory Management System for Alekos' Corner General Merchandise."** This project is being developed as a requirement for the subject **IT083 – IT Project Management**.

The proposed system aims to provide Alekos' Corner General Merchandise with a customized Point-of-Sale and inventory management system that can help improve sales transaction processing, inventory monitoring, product and price management, and sales reporting. In developing the system, we intend to gather relevant information about the business's existing processes and requirements to ensure that the proposed system is aligned with the client's operational needs.

In this regard, we respectfully request your approval to conduct the necessary project activities, including the gathering and documentation of information relevant to the development of the proposed system. Any information obtained for this project will be used solely for academic and system development purposes and will be handled with appropriate confidentiality.

We sincerely appreciate your time, guidance, and consideration regarding this project.

Thank you very much.

Respectfully yours,

Aldave, Resty S.
Bagahansol, Shan Lester S.
Ramal, John Wilfred Y.
Llaguno, Clary Mae F.
Student Researchers/Proponents

APPENDIX C

INTERVIEW GUIDE

This interview guide is designed to gather detailed information about the current operations, sales processes, inventory management, record-keeping practices, and challenges experienced by Alekos' Corner General Merchandise. The interview will supplement the Client Business Roles & Process Flow Survey by providing opportunities for the client to explain existing procedures, identify specific operational difficulties, and describe the features and requirements expected from the proposed Managing Sentry system. The information gathered will be used as a basis for defining the system requirements and designing the system according to the needs of the business.

The following questions will guide the interview and focus on the client's business roles, current processes, operational challenges, and system requirements.

APPENDIX D

QUESTIONNAIRE

Client Business Roles & Process Flow Survey

Purpose: To understand the client's current business roles and workflow, in order to design the system's user roles and modules accurately.

Client / Business Name: _____

Respondent Name & Position: _____

Date: _____

A. Business Roles

#	Question	Notes / Answer
1	Who are all the people involved in the day-to-day operations (owner, staff, cashier, client, etc.), and what does each one do?	
2	Who has the authority to approve, reject, or edit records/transactions?	
3	Are there different access levels needed (e.g., Admin sees everything, Staff sees limited data)?	

B. Process Flow

#	Question	Notes / Answer
4	Can you walk us through a typical transaction from start to finish? What are the exact steps?	
5	What forms or paper documents are currently used, and what information do they capture?	
6	Where do records get stored right now, and who can access them?	
7	What happens when there's an error or a record needs to be corrected?	

C. Pain Points & Requirements

#	Question	Notes / Answer
8	What part of the current manual process takes the most time or causes the most mistakes?	
9	Are there reports you need to generate regularly (daily, weekly, monthly)? What information do those reports need?	
10	Is there a busy season or peak time where the manual process becomes a bigger problem?	

Additional Notes:

_____

_____

APPENDIX E

SOFTWARE REQUIREMENT SPECIFICATIONS

E.1 Introduction

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements of Managing Sentry, a Point-of-Sale (POS) and Inventory Management System developed for Alekos' Corner General Merchandise. The system is designed to improve sales transaction processing, inventory monitoring, product and pricing management, sales reporting, and recordkeeping through an Android-based POS application and a web-based administration portal. It also supports offline transaction recording, automatic inventory updates, low-stock notifications, and product expiration alerts. This document specifies the system features, user roles, operational requirements, limitations, and performance expectations that will guide the development, testing, and implementation of Managing Sentry.

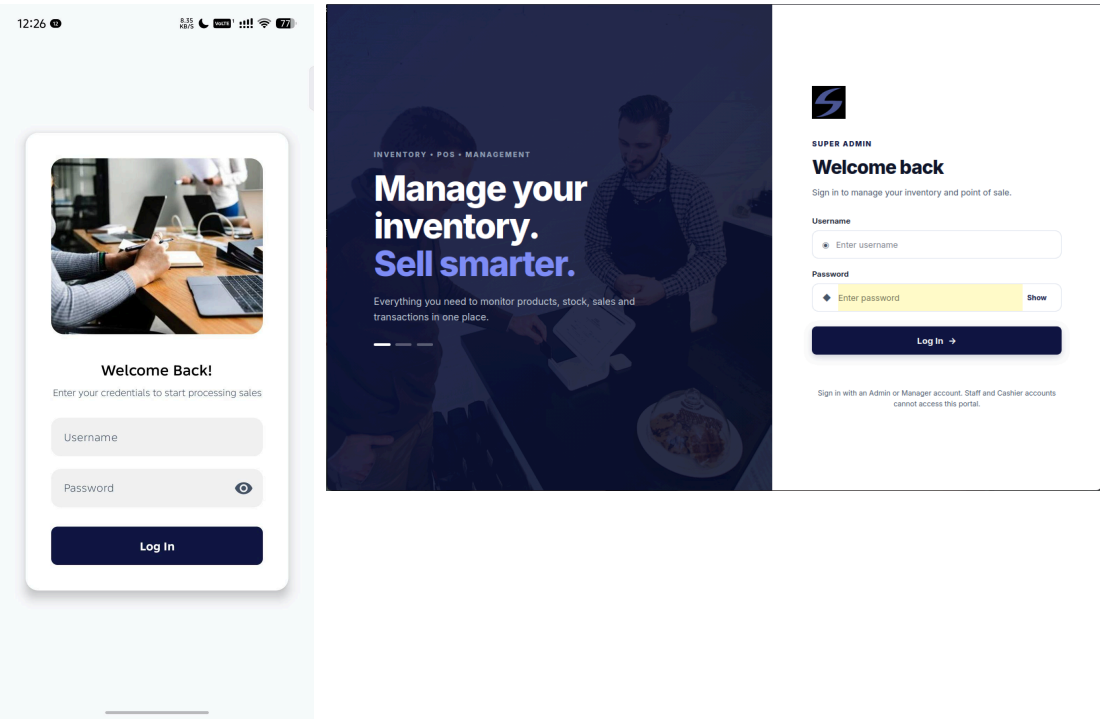
E.2 Functional Requirements

Functional requirements describe the specific operations and features that Managing Sentry must perform.

E.2.1 User Authentication and Account Management

1. The system shall allow authorized users to log in using their registered credentials.
2. The system shall provide access according to the user's assigned role and permissions.
3. The system shall restrict administrative functions to authorized users.
4. The system shall allow authorized administrators to manage user accounts and their assigned roles.

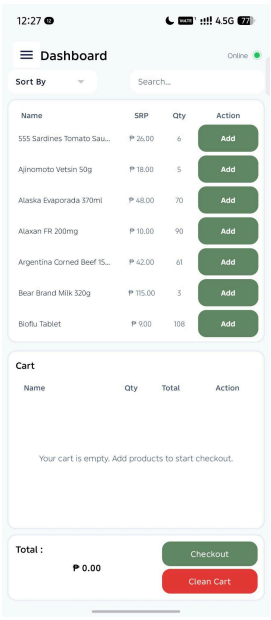
5. The system shall prevent unauthorized users from accessing restricted system functions.

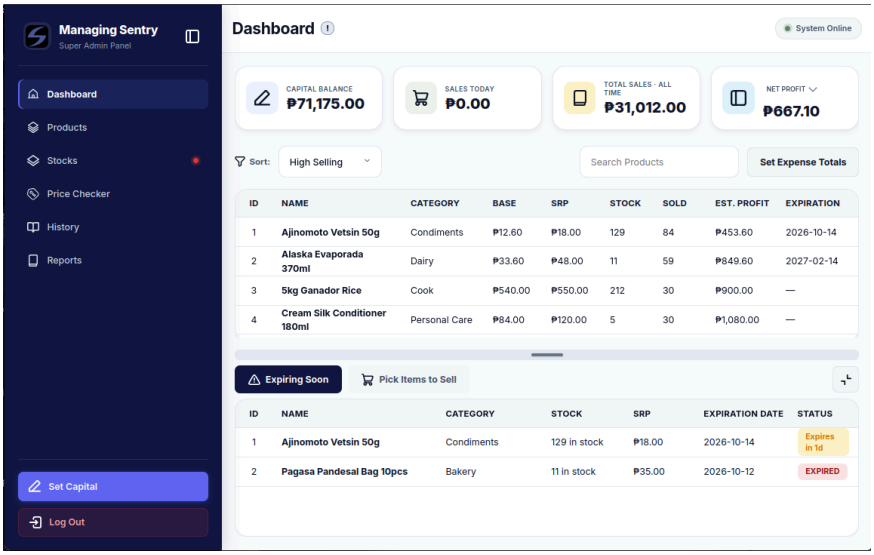


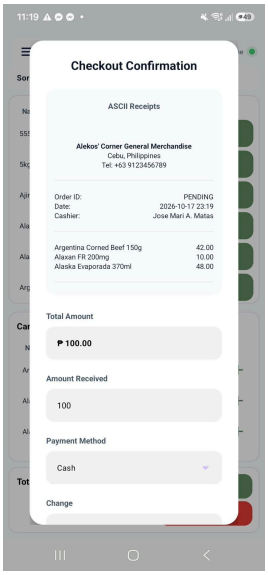
E.2.2 Point-of-Sale (POS) and Transaction Processing

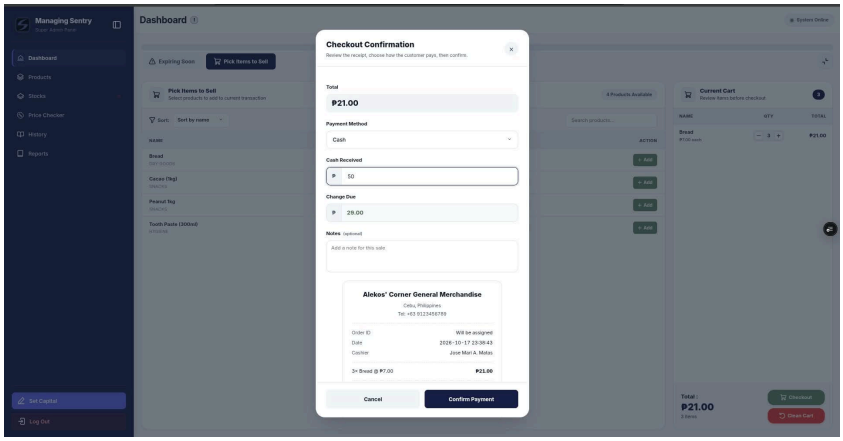
1. The Android POS application shall allow staff to manually select products during checkout.
2. The system shall retrieve the recorded product price and use it to calculate the transaction total.
3. The system shall calculate the total amount based on the selected products, quantities, and applicable recorded prices.
4. The system shall allow staff to record cash-based or manually recorded sales transactions.
5. The system shall save completed transaction details, including the products purchased, quantities, prices, total amount, and transaction date and time.

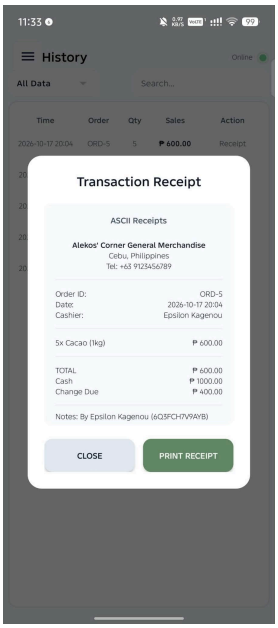
6. The system shall update inventory quantities after a completed sale has been recorded.
7. The system shall support offline transaction recording when an internet connection is unavailable, subject to the application's implemented offline capabilities.
8. The system shall synchronize locally stored transactions with the central database when connectivity becomes available.
9. The system shall provide a mechanism to track synchronization status and handle failed synchronization attempts.
10. The system shall not automatically apply senior citizen or Persons with Disabilities (PWD) discounts. Any applicable discounts must be calculated and applied manually by authorized staff, and the system's recorded transaction total must reflect the amount actually collected.

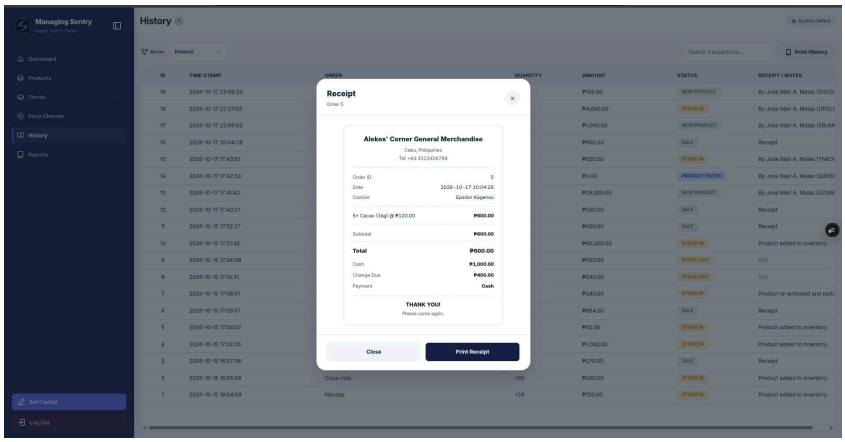








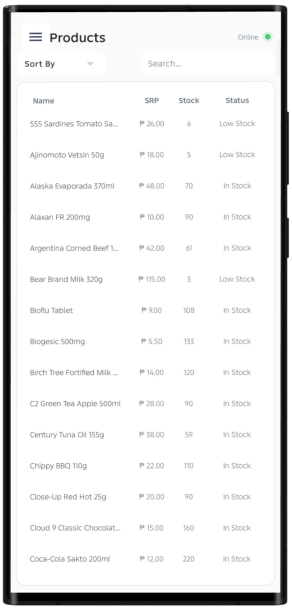


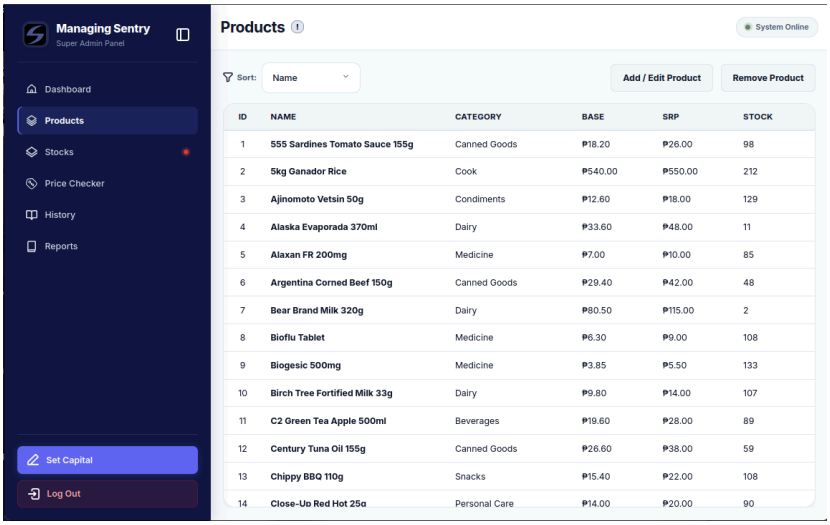


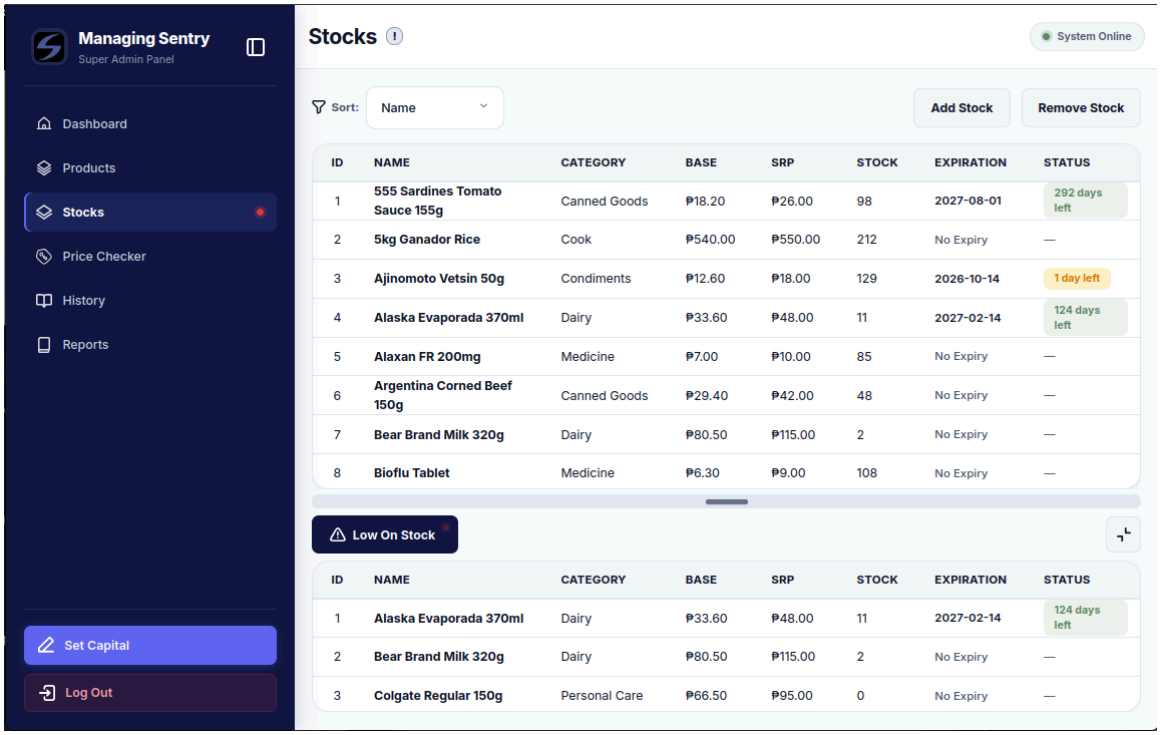
E.2.3 Product and Inventory Management

1. The system shall allow authorized users to add new products to the inventory.
2. The system shall allow authorized users to view and update product information, including product names, prices, quantities, and expiration dates where applicable.

3. The system shall allow authorized users to manage product prices and stock quantities.
4. The system shall automatically deduct sold quantities from the recorded inventory after completed transactions.
5. The system shall allow authorized users to record restocking activities and the quantities added to inventory.
6. The system shall maintain inventory history records for restocking activities and other supported stock adjustments.
7. The system shall display current recorded stock quantities to support inventory monitoring.
8. The system shall allow authorized users to review inventory records to identify discrepancies and correct them when necessary.

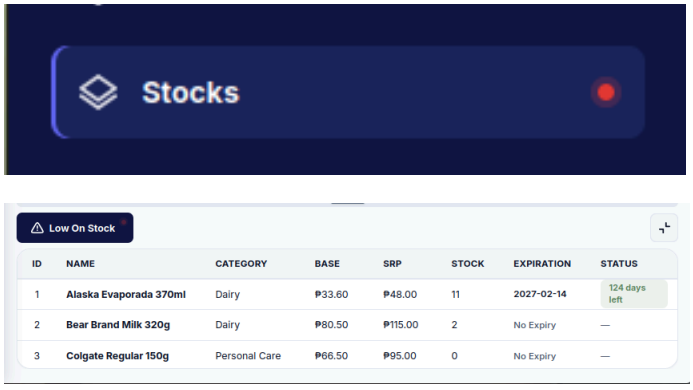






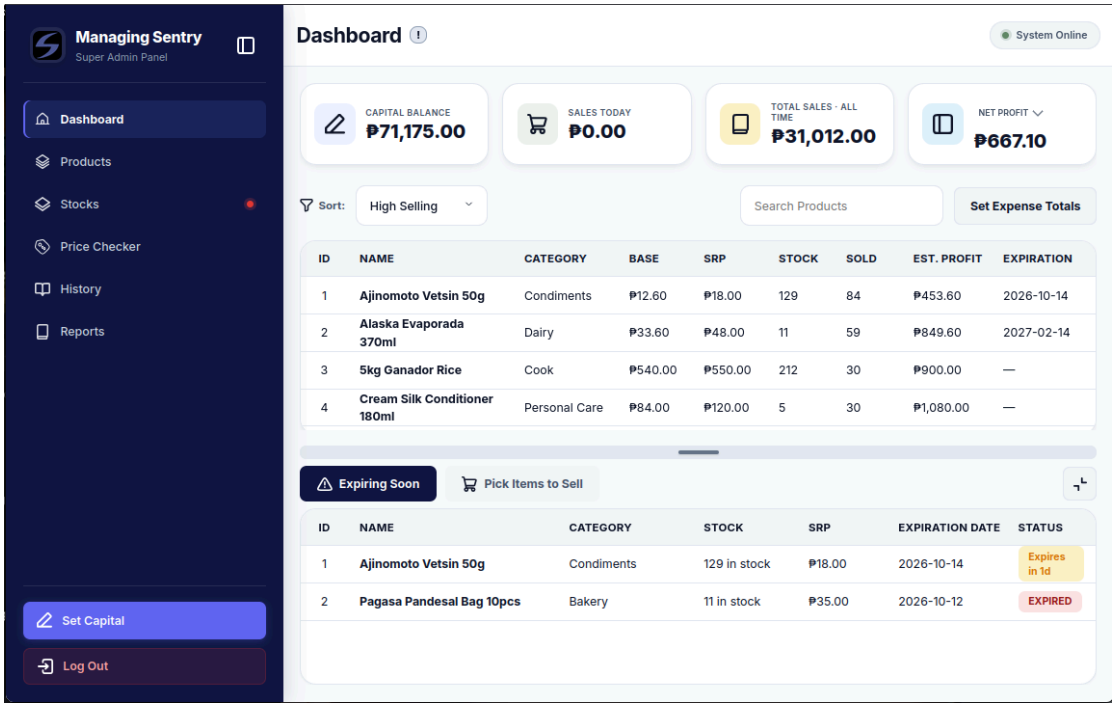
E.2.4 Low-Stock Notifications

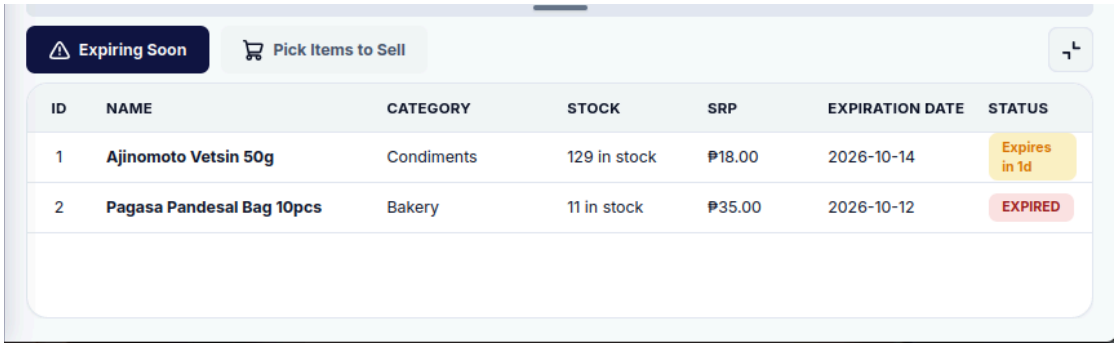
1. The system shall monitor product quantities based on the recorded inventory levels.
2. The system shall identify products whose stock quantities reach or fall below the configured minimum-stock threshold.
3. The system shall use a minimum-stock threshold equivalent to 20% of the product's configured stock quantity, according to the system's defined calculation.
4. The web-based Super Admin interface shall display a red blinking indicator for products that meet the low-stock condition.
5. The system shall allow the Super Admin to review flagged products and determine whether restocking is necessary.



E.2.5 Product Expiration Monitoring

1. The system shall store expiration dates for products with applicable expiration information.
2. The system shall check recorded expiration dates to identify products approaching expiration.
3. The system shall display an expiration alert when a product is within seven days of its recorded expiration date.
4. The system shall allow authorized administrators to review products identified by expiration alerts and take appropriate inventory actions.

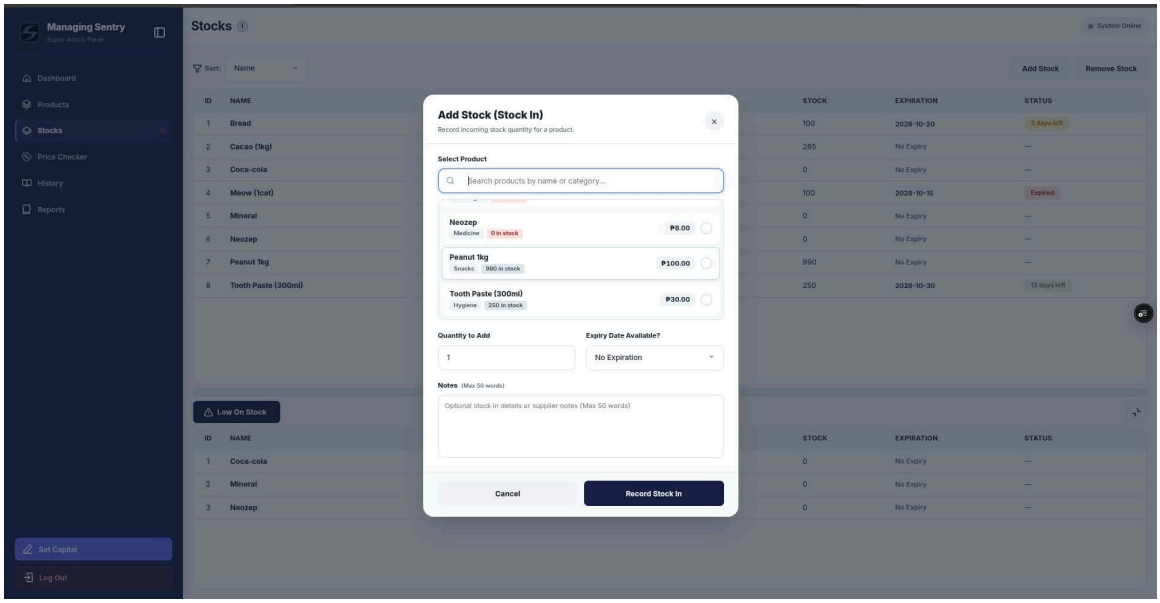




E.2.6 Restocking and Inventory History

1. The system shall allow the Admin or the manager to record inventory restocking activities.
2. The system shall update the recorded stock quantity after a restocking entry is saved.
3. The system shall record relevant restocking details, including the product, quantity added, date and time, and authorized user where supported.

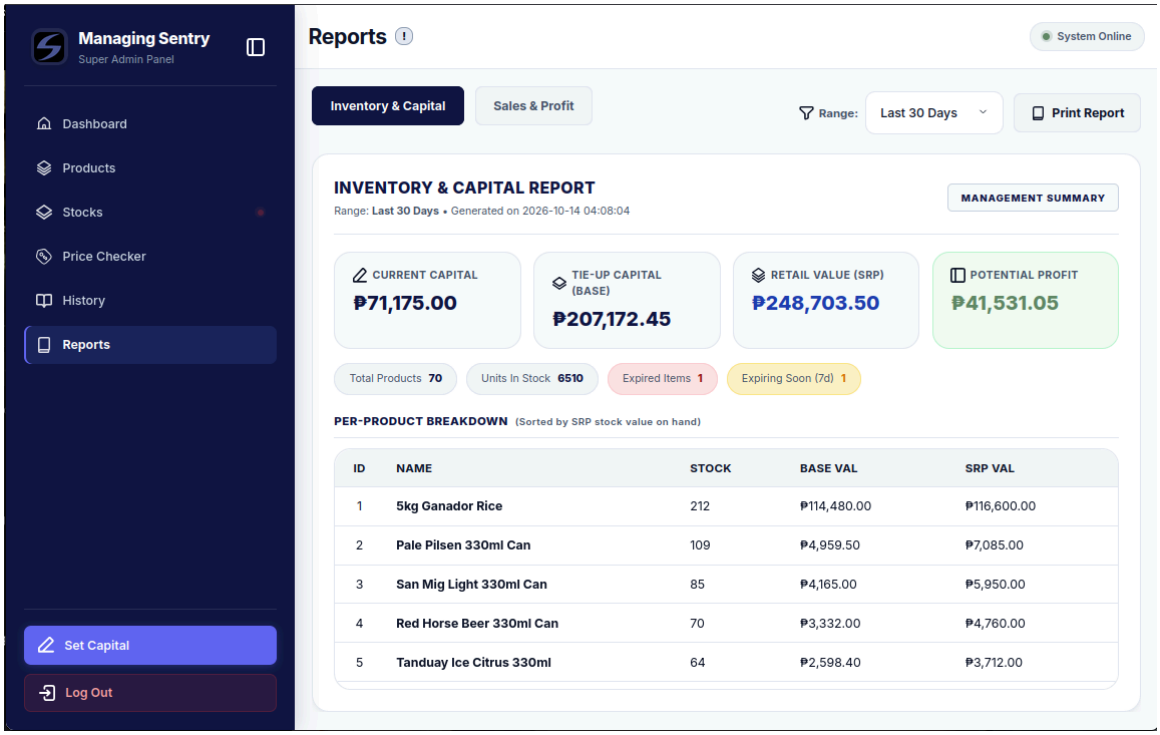
4. The system shall allow authorized users to include notes explaining restocking activities or stock adjustments.
5. The system shall provide access to recorded inventory history for review and monitoring.



E.2.7 Sales Reports and Business Monitoring

1. The web-based administration portal shall provide sales reports for daily, weekly, and monthly periods.
2. The system shall calculate and display total recorded sales revenue for the selected reporting period.
3. The system shall provide a summary of products sold and their recorded quantities.
4. The system shall identify best-selling products based on recorded sales data.
5. The system shall allow authorized administrators to review sales records to support business monitoring and decision-making.

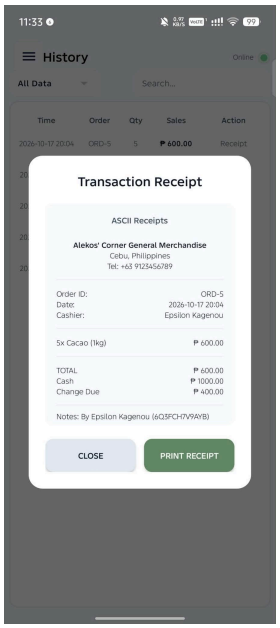
6. The system shall generate reports using available transaction records and shall account for synchronization status when displaying data from offline transactions.



E.2.8 Customer Receipt Issuance

1. The initial system shall record transaction details without requiring an integrated receipt printer.
2. If the client requests receipt printing in the future, the development team shall evaluate the required hardware, software modifications, project scope, and additional costs.
3. The receipt issuance feature may be implemented only after the applicable legal, tax, and business requirements have been verified and the necessary permits or authorizations have been obtained where required.

4. If the feature is implemented, the system shall generate receipt information based on the recorded transaction details and provide printing functionality compatible with the selected printer.
5. The system shall preserve the recorded transaction if receipt printing fails and allow authorized staff to retrieve or reprint the receipt when appropriate.
6. Receipt printing and any applicable official receipt functionality shall be subject to successful testing and verification of applicable requirements.



E.3 Non-Functional Requirements

Non-functional requirements describe the expected quality, security, reliability, and operational characteristics of Managing Sentry.

E.3.1 Usability

1. The Android POS application shall provide a clear and understandable interface for staff processing sales.
2. The web-based administration portal shall organize inventory, alerts, and reports in a manner that allows authorized users to locate relevant information easily.
3. Labels, buttons, and messages shall use clear and consistent wording.
4. The system shall provide understandable feedback for completed operations, validation errors, and synchronization problems.

E.3.2 Performance

1. The system shall calculate transaction totals using the selected products, quantities, and recorded prices.
2. The system shall update inventory records after a completed transaction has been successfully recorded.
3. The system shall retrieve inventory information and generate reports within a reasonable response time under normal operating conditions.
4. The system shall process offline transactions without requiring an active internet connection for the supported offline operations.

E.3.3 Reliability and Availability

1. The system shall preserve successfully recorded transactions during normal operation.
2. The system shall provide a mechanism to identify and recover from failed synchronization attempts.
3. The system shall prevent duplicate transaction records during synchronization through appropriate validation and duplicate-prevention mechanisms.

4. The system shall maintain consistent transaction and inventory records when supported operations are completed successfully.
5. The system shall provide backup and recovery procedures appropriate to the database environment.

E.3.4 Security

1. The system shall require authentication before granting access to restricted functions.
2. The system shall enforce role-based permissions for staff and administrative users.
3. The system shall protect user credentials and sensitive business records through appropriate security controls.
4. The system shall validate input data to reduce invalid entries and unauthorized modifications.
5. The system shall restrict product management, inventory adjustments, user management, and administrative reporting to authorized roles as defined by the system's permissions.

E.3.5 Data Integrity

1. The system shall validate required product, transaction, and inventory information before saving records.
2. The system shall ensure that transaction totals are calculated consistently using the recorded prices and quantities.
3. The system shall maintain relationships between sales transactions and their corresponding product details.
4. The system shall apply inventory changes only when the corresponding transaction or stock adjustment has been successfully recorded.
5. The system shall preserve relevant transaction and inventory history to support record review and discrepancy investigation.

E.3.6 Maintainability

1. The system shall organize its application components, API endpoints, and database structures to support future maintenance.
2. The development team shall document important system functions, database structures, and integration procedures.
3. The system shall allow authorized developers to correct defects and implement approved enhancements without unnecessarily disrupting unrelated features.
4. Changes to existing functions shall undergo appropriate testing before deployment.

E.3.7 Compatibility

1. The Android POS application shall operate on Android devices that meet its documented software and hardware requirements.
2. The web-based administration portal shall operate on supported modern web browsers.
3. The system's application components shall communicate through their configured API interfaces.
4. Any future receipt printer integration shall be tested against the selected device and its supported connection method.

E.4 System Limitations

1. Managing Sentry is designed specifically for the operational needs of Alekos’ Corner General Merchandise and is not initially intended to support multiple independent retail businesses.
2. Product selection during checkout is performed manually. Barcode scanners, weighing equipment, and other specialized hardware are not included in the initial system scope.

3. The system focuses on in-store, cash-based, or manually recorded transactions. It does not integrate with external card terminals, online payment services, or banking APIs.
4. The system does not automatically calculate or apply senior citizen and PWD discounts. Authorized staff must calculate and apply any applicable discounts manually, and the transaction must be recorded using the appropriate final amount.
5. Customer receipt printing is not included in the initial system scope. It may be considered as a future enhancement after the applicable requirements, authorizations, hardware needs, and costs have been evaluated.
6. Offline functionality is limited to the operations implemented for local transaction recording and synchronization. Features that require access to the central database or internet-based services may be unavailable while offline.
7. The accuracy of sales reports, inventory records, low-stock alerts, and expiration alerts depends on the accuracy and completeness of the information entered and recorded in the system.

E.5 Summary

The functional and non-functional requirements defined in this document provide the basis for the development and evaluation of Managing Sentry. The functional requirements specify the operations the system must support, while the non-functional requirements define its expected usability, performance, reliability, security, data integrity, maintainability, and compatibility. The documented limitations establish the boundaries of the initial implementation and identify features, such as customer receipt printing and automated senior citizen or PWD discount application, that are not included in the initial system scope. These requirements will guide system development, functional testing, and validation against the operational needs of Alekos' Corner General Merchandise.

Category	Development Specifications	Deployment / Client Specifications
Hardware Capabilities	• **Processor:** Intel Core i3 (11th Gen or higher) / Equivalent AMD Ryzen • **RAM:** Minimum 8GB • **Storage:** SSD Storage	• **Devices:** Desktop/Laptop, Tablet, or Smartphone • **Processor / RAM:** Standard consumer devices with a minimum of 2GB–4GB RAM and stable display resolution (responsive web layout).
Operating System	• Windows 10+	• OS Independent (Cross-platform compatibility via browser)

	• Fedora Linux (or Ubuntu)	
Software & Tools	• VS Code • Git & GitHub • Figma (UI/UX Design)	• Modern Web Browser (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari)
Database & Hosting	• PostgreSQL / MySQL • DBeaver (Database Management) • Local Instance Environment	• Cloud Database Hosting (Supabase / Render / Vercel Host) accessed securely via web browser

CURRICULUM VITAE



CONTACT INFORMATION

Full Name: John Wilfred Yu Rama

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Phone Number:09205060***

Email Address: joyu.rama.swu@phinmaed.com

PERSONAL INFORMATION

Date of Birth: July 20, 2006

Age:20

Citizenship/Nationality: Filipino

Gender: Male

Marital Status: Single

EDUCATIONAL BACKGROUND

Educational Level

Primary: Kiddiehaus Of Learning (2010-2019)

Secondary: Abellana National School (2019-2023)

: SouthWestern University PHIMA (2023-2025)

Tertiary Education : SouthWestern University PHIMA (2025-Present)

Certifications & Accreditations

Dean's Lister 2025-2026

Java Programming for Beginners 2026

Responsive Web Design 2026

Python Essentials 2025

CAREER OBJECTIVES

"A better future with technology using knowledge."

CURRICULUM VITAE

CONTACT INFORMATION



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Current Address: Andress Abellana Ext. Cebu City

Phone Number: 09476962309

Email Address: shse.bagahansol.swu@phinmaed.com

PERSONAL INFORMATION

Date of Birth: 08/01/2004

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Citizenship/Nationality: Filipino

Gender: Male

Marital Status: Single

EDUCATIONAL BACKGROUND

Educational Level

Primary:Guadalupe Elementary School (2010-2016)

Secondary: Ramon Duterte Memorial National High School (2017-2020)

: Southwestern University PHINMA (2023-2025)

Tertiary Education : Southwestern University PHINMA (2025 - Present)

Certifications & Accreditations

Cisco Networking Academy (2024)

freeCodeCamp (2024)

CAREER OBJECTIVES

“To use technology to build better tools and help improve the local community”

CURRICULUM VITAE

CONTACT INFORMATION



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Phone Number: 9604439777

Email Address: resi.aldave.swu@phinmaed.com

PERSONAL INFORMATION

Date of Birth: May 17, 2007

Age: 19

Citizenship/Nationality: Filipino

Gender: Male

Marital Status: Single

EDUCATIONAL BACKGROUND

Educational Level

Primary: Upland Elementary School (2013 - 2019)

Secondary: Carcar Academy Technical School Inc. (2019 - 2025)

:

Tertiary Education : Southwestern University (2025 - Present)

Certifications & Accreditations

- Cisco Python Essentials 1 2025
- Java Programming for Beginners 2026
- Responsive Web Design 2026
- Cisco Networking Basics & Essentials 2026

CAREER OBJECTIVES

“Make life better with the digital world and create a safe environment”

CURRICULUM VITAE



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PERSONAL INFORMATION

Date of Birth: February 21, 2007

Age: 19

Citizenship/Nationality: Filipino

Gender: Female

Marital Status: Single

EDUCATIONAL BACKGROUND

Educational Level

Primary: University of Cebu Main Campus (2013 - 2019)

Secondary: University of Cebu Main Campus (2019 - 2023)

: University of Cebu Pri (2023 - 2025)

Tertiary Education: Cebu Institute of Technology (2025)

: Southwestern University (2025 - Present)

Certifications & Accreditations

CAREER OBJECTIVES

“To protect data privacy and critical infrastructure, treating every line of defense as a vital shield for people's digital lives and livelihoods."